AP CALCULUS AB - UNIT 1

1.6 Determining Limits Using Algebraic Manipulation

Lesson Overview

- Use substitution as a diagnostic rather than a final answer when it produces an indeterminate form.
- Select factoring, conjugates, common denominators, trigonometric identities, or side-specific algebra efficiently.
- Justify cancellation through equality on a punctured neighborhood.
- Use the standard trigonometric limits in radians.

Key Knowledge and Vocabulary

Diagnostic substitution. A finite value completes the problem; $0/0$ signals that algebra may reveal hidden local behavior.

Punctured equivalence. Expressions that agree for all nearby $x \neq c$ have the same limit at c.

Factoring. Cancel common multiplicative factors, never terms separated by addition.

Conjugates. Use $(A - B)(A + B) = A^2 - B^2$ to expose a factor.

Trig anchors. $\lim_{u\to 0} \frac{\sin u}{u} = 1$ and $\lim_{u\to 0} \frac{1-\cos u}{u} = 0$ in radians.

Explanation and Strategy

Step 1. First substitute and classify the result. Then choose the shortest structure-preserving rewrite.
Step 2. State the excluded target when canceling a factor; the simplified expression is used only for nearby inputs.
Step 3. For absolute value and piecewise expressions, simplify separately on each side before comparing.

Solved Examples

Worked Example: Factoring a removable form

Evaluate $\displaystyle\lim_{x\to 5} \frac{x^2 - 25}{x - 5}$.

Solution. Factor: $\frac{(x-5)(x+5)}{x-5} = x + 5$ for $x \neq 5$. Therefore the limit is 10.

Worked Example: Rationalizing a numerator

Evaluate $\displaystyle\lim_{x\to 4} \frac{\sqrt{x+5} - 3}{x - 4}$.

Solution. Multiply by the conjugate. The expression becomes $\frac{1}{\sqrt{x+5}+3}$ for $x \neq 4$, so the limit is $1/6$.

Worked Example: Trigonometric scaling

Evaluate $\displaystyle\lim_{x\to 0} \frac{\sin(7x)}{x}$.

Solution. Write $7\frac{\sin(7x)}{7x}$. Since $7x \to 0$, the limit is 7.

Leveled Practice

Easy Level

Foundational fluency. Focus on notation, definitions, and direct procedures.

1. Evaluate $\lim_{x\to 5} \frac{x^2 - 25}{x - 5}$.

2. Evaluate $\lim_{x\to 2} \frac{x^2 + x - 6}{x - 2}$.

3. Evaluate $\lim_{x\to 4} \frac{\sqrt{x+5} - 3}{x - 4}$.

4. Evaluate $\lim_{x\to 0} \frac{\sin(7x)}{x}$.

Medium Level

Apply the idea in one or two connected steps.

5. Evaluate $\lim_{x\to 2} \frac{x^3 - 8}{x - 2}$.

6. Evaluate $\lim_{x\to 4} \frac{\frac{1}{x} - \frac{1}{4}}{x - 4}$.

7. Evaluate $\lim_{x\to 9} \frac{x - 9}{\sqrt{x} - 3}$.

8. Find $\lim_{x\to -2^-} \frac{|x+2|}{x+2}$ and $\lim_{x\to -2^+} \frac{|x+2|}{x+2}$.

Hard Level

Connect representations, conditions, and justification.

9. Evaluate $\lim\limits_{x\to 2} \dfrac{x^4-16}{x-2}$.

10. Evaluate $\lim\limits_{x\to 1} \dfrac{\sqrt{2x+7}-3}{x-1}$.

11. Evaluate $\lim\limits_{x\to 0} \dfrac{1-\cos(3x)}{x^2}$.

12. Evaluate $\lim\limits_{x\to 0} \dfrac{\sin(5x)}{\sin(2x)}$.

Difficult Level

Use nonroutine structure and precise mathematical reasoning.

13. Evaluate $\lim\limits_{x\to 2} \dfrac{x^3-8}{\sqrt{x+2}-2}$.

14. Evaluate $\lim_{x \to 1} \dfrac{\sqrt[3]{x+7} - 2}{x - 1}$.

15. Evaluate $\lim_{x \to 3} \dfrac{\frac{1}{x+2} - \frac{1}{5}}{x - 3}$.

Challenge Level

Synthesize, construct, generalize, or prove.

16. Prove that for positive integer n, $\lim_{x \to a} \dfrac{x^n - a^n}{x - a} = na^{n-1}$.

17. Evaluate $\lim_{x \to 0} \dfrac{\sqrt{1+3x} - \sqrt{1-2x}}{x}$.

AP-Style Practice

Multiple-Choice Practice – No Calculator

Choose the best answer. Use exact values unless a decimal approximation is requested.

1. $\lim_{x\to 3} \dfrac{x^2-9}{x-3}$ equals

(A) 0
(B) 3
(C) 6
(D) DNE

2. $\lim_{x\to 0} \dfrac{\sin(4x)}{3x}$ equals

(A) $\frac{3}{4}$
(B) $\frac{4}{3}$
(C) 3
(D) 4

3. Which first move is most efficient for $\lim_{x\to 9} \frac{\sqrt{x}-3}{x-9}$?

(A) Factor a quadratic
(B) Multiply by the conjugate
(C) Use a sign chart
(D) Create a table only

Multiple-Choice Practice – Calculator Required

Choose the best answer. Use exact values unless a decimal approximation is requested.

4. A calculator suggests $\lim_{x\to 0} \frac{e^{2x}-1}{x} \approx 2$. Which table best supports the conclusion?

(A) Only $x = 0$
(B) Inputs near 0 from both sides with increasing precision
(C) Only positive integers
(D) Only $x = 1, 2, 3$

5. Numerical values of $\frac{1-\cos x}{x^2}$ near 0 approach

(A) 0
(B) $\frac{1}{2}$
(C) 1
(D) $+\infty$

Free-Response Practice – No Calculator

Consider $F(x) = \dfrac{x^3+8}{x+2}$.

(a) Explain why direct substitution at $x = -2$ is inconclusive for the displayed quotient. [1 points]

(b) Find $\lim_{x\to -2} F(x)$ algebraically. [3 points]

(c) Describe the graph of F relative to the simplified polynomial near $x = -2$. [2 points]

Free-Response Practice – Calculator Required

Use a calculator to investigate $H(x) = \dfrac{\sqrt{1+5x}-1}{x}$ near $x = 0$.

(a) Estimate the limit numerically to three decimal places. [2 points]

(b) Confirm the estimate algebraically. [3 points]

(c) Explain why very small decimal inputs may produce unstable calculator values in the original form. [2 points]